**Shared Nominal Models for Fleet-Wide Condition Monitoring**

Independent Operational-Radius Convergence Across Heterogeneous Assets

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**Abstract**

**Objectives:** Transportation operators deploying condition monitoring across large fleets of heterogeneous assets rarely have fault labels, time for per-asset representation learning, or long field calibration campaigns. This paper addresses how a single nominal representation can be deployed fleet-wide on day one, without retraining a model or tuning a threshold for every asset.

**Methods:** A canonical nominal representation, a Conv1D autoencoder, is trained once offline on simulated healthy vibration signals and transferred unchanged to every asset. Each asset then estimates its operational radius, an empirical residual quantile, from a short stream of unlabeled local observations, without retraining the shared model. The method is evaluated on four Case Western Reserve University (CWRU) bearing conditions, spanning healthy, ball-fault, inner-race, and outer-race windows, against an oracle calibrated on each condition's complete healthy dataset. Three calibration strategies are compared: direct threshold transfer, a complete-data oracle radius, and the proposed online radius estimated from only the first 100 healthy windows.

**Findings:** Direct transfer of the absolute simulator threshold produced complete false alarms in the field, confirming that the representation, not its absolute score scale, transfers. Using only a 100-window online calibration, the shared representation reached an average healthy false-alarm rate of 3.09%, ball-fault detection of 99.69%, and 100% detection of inner- and outer-race faults, with the radius converging after 51.0 ± 12.6 windows, approaching the asset-specific reference.

**Novelty:** The contribution is a fleet-deployment methodology, not a new anomaly-detection architecture: it separates representation learning, performed once and shared fleet-wide, from decision-boundary calibration, performed independently by each asset.

**Practical Applications:** A shared representation can be developed and maintained centrally, for example by an original equipment manufacturer, and distributed unchanged across an operator's fleet, while each installed asset requires only seconds of unsupervised local calibration and lightweight edge inference, removing the need for fault labels or per-asset model training during rollout.

**Keywords:** condition monitoring, condition-based maintenance, transportation asset management, fleet deployment, anomaly detection, cold start, local calibration, edge monitoring

# 1 Introduction

Modern transportation systems increasingly rely on condition-monitoring technologies to supervise large populations of operational assets, including rolling-element bearings, axleboxes, traction motors, gearboxes, pumps, actuators, vehicles, and spatially indexed infrastructure segments Jardine et al. 2006; Zhao et al. 2019. Although machine-learning research has produced increasingly capable anomaly detectors Chandola et al. 2009; Zhao et al. 2019, the deployment problem remains largely unresolved: a detector that performs well for one asset under one calibrated operating condition does not automatically scale to hundreds or thousands of independently installed units.

At fleet rollout, several constraints arise simultaneously. First, the target fleet contains few or no representative fault examples. Second, waiting months or years for failures to accumulate is operationally unacceptable. Third, installation-specific factors such as sensor coupling, mounting, operating point, and environmental noise alter the numerical scale of anomaly scores between assets Randall and Antoni 2011. Fourth, training, calibrating, validating, and maintaining a separate representation model for every physical unit creates an operational and organizational burden that grows directly with fleet size.

The practical question is therefore not simply

> *How can an anomalous condition be detected?*

but rather

> *How can one nominal representation be deployed on day one across a heterogeneous fleet while retaining performance close to an ideal asset-specific solution, without per-asset representation learning or manual threshold tuning?*

This paper addresses that question by separating representation engineering from deployment calibration. A canonical nominal representation is learned once, offline, from simulated healthy observations and is distributed unchanged across the fleet. Each deployed asset estimates only its own operational radius from a short stream of unlabeled local observations. The shared model is not retrained, fine-tuned, or adapted in the field.

From a broader perspective, this separation can be interpreted as an inverse problem. Representation learning is performed once to recover a canonical nominal representation from observations, whereas deployment requires each asset to solve only a lightweight local inverse calibration problem by identifying its own operational radius around that fixed representation. This interpretation follows the inverse-problem and quotient-space formulations introduced in our previous work Di Santi 2026b; Di Santi 2026a, while the present paper focuses on their engineering realization for scalable fleet deployment.

For asset $k$, let $\tau_k(t)$ denote the progressively estimated operational radius. Under stable nominal operation, the proposed calibration procedure is expected to satisfy

$$\tau_k(t) \to \tau_k^\star,$$

where $\tau_k^\star$ is an asset-specific acceptance boundary around the fixed shared representation. Once convergence is established, the radius is frozen or tightly constrained so that subsequent degradation cannot redefine itself as nominal through unrestricted adaptation.

The distinction between the transferable representation and the local decision boundary is essential. A reconstruction-error threshold learned in simulation is expressed in the numerical scale induced by the simulator, its preprocessing, and its residual distribution. Applying that absolute value directly to a field installation may therefore fail even when the underlying representation remains diagnostically useful. The proposed method transfers neither an absolute threshold nor an asset-specific model. Instead, it transfers the canonical nominal representation and a fleet-wide target nominal coverage, while each asset estimates the empirical residual quantile that realizes that coverage locally.

The transferable object is therefore not an anomaly threshold, but a canonical nominal representation. Deployment consists only of identifying the local operational radius around that shared representation.

This formulation also changes the relevant experimental comparison. The ideal reference is not another globally shared detector, but an operationally unrealistic condition in which the complete healthy distribution of every asset is available in advance and a separate detector can be trained and calibrated for each one. The central question is therefore

> *How much detection performance is lost when many asset-specific models are replaced by one simulator-trained representation and a few seconds of local calibration?*

We answer this question using four operating conditions from the Case Western Reserve University (CWRU) bearing benchmark Smith and Randall 2015. The asset-specific reference methods are trained and calibrated using the complete healthy dataset of each condition. In contrast, the proposed method uses no field observation for representation learning and estimates its local operational radius from only the first 100 healthy windows, corresponding to approximately ten seconds of signal at the evaluated window duration.

The experimental results show that the cost of scalable deployment is small. Asset-specific autoencoders reach 100% detection across the evaluated ball-, inner-race-, and outer-race-fault families. The unchanged simulator-trained shared model, combined with online local-radius estimation, reaches an average ball-fault detection rate of 99.69% and 100% detection of inner- and outer-race faults. The operational radius converges after $51.0 \pm 12.6$ windows on average. Direct transfer of the absolute simulator threshold, however, produces 100% healthy false alarms, confirming that the representation transfers while its absolute score scale does not.

The contribution of this paper is therefore a deployment methodology rather than a new neural architecture. Although a Conv1D autoencoder is used as the experimental representation, the deployment principle is representation-agnostic. The principal contributions are:

1. a fleet-wide architecture that separates one shared canonical nominal representation from independently calibrated local operational radii;

2. a direct simulation-to-field experiment in which the representation is transferred unchanged and no field data are used for representation learning;

3. an online empirical-quantile calibration procedure that estimates an asset-specific operational radius from a short unlabeled warm-up;

4. a convergence criterion that separates finite cold-start calibration from subsequent degradation monitoring;

5. a comparison against ideal asset-specific detectors trained and calibrated with the complete healthy dataset of every evaluated condition; and

6. an edge-oriented deployment model in which the representation may be produced and maintained centrally—for example by an original equipment manufacturer—while each deployed unit performs only lightweight local calibration.

The methodology is intentionally independent of asset type and sensing modality. It requires only a nominal representation producing a scalar anomaly score, repeated observations from each operational unit, and a local procedure for estimating the acceptance boundary. Public rolling-element bearing data are used here as the first controlled validation domain rather than as a limitation of the proposed methodology.

The remainder of the paper is organized as follows. Section 2 reviews condition monitoring, one-class anomaly detection, adaptive calibration, and fleet deployment. Section 3 defines the shared representation and local operational-radius estimation. Section 4 presents the experimental protocol. Section 5 reports the deployment-oriented results, and Section 6 discusses industrial implications, limitations, and future work. Section 8 summarizes the underlying mathematical interpretation.

## 2 Related Work

Condition-based maintenance (CBM) has become a central component of modern transportation asset management, with condition monitoring providing the operational evidence required to supervise large fleets of heterogeneous assets Jardine et al. 2006; Chandola et al. 2009; Zhao et al. 2019. Although the sensing modality depends on the application—including vibration, acoustic emission, electrical current, temperature, strain, or other physical measurements—the computational objective is fundamentally the same: to characterize nominal operation and detect persistent departures that may indicate degradation.

Most machine-learning approaches formulate this problem as supervised fault classification, assuming representative fault examples are available during training. While effective on benchmark datasets, supervised methods require fault labels and generally assume that deployment conditions remain close to those observed during model development.

One-class and novelty-detection methods, including one-class SVM Schölkopf et al. 1999, neural-network novelty detection Bishop 1994, and autoencoder-based anomaly detection Hinton and Salakhutdinov 2006; Sakurada and Yairi 2014, reduce the dependence on fault labels by learning only nominal behaviour. More recently, representation-learning approaches have substantially improved anomaly detection by constructing latent spaces in which healthy operation occupies a compact region while degraded behaviour becomes more easily separable.

However, the overwhelming majority of existing work focuses on learning better anomaly detectors. Comparatively little attention has been paid to the deployment problem itself: how a single nominal representation should be initialized, calibrated, and maintained across a heterogeneous fleet of independently operating assets.

Adaptive thresholds are frequently employed in streaming anomaly detection to compensate for changes in operating conditions. However, unrestricted adaptation introduces an important operational risk: persistent degradation may progressively redefine itself as nominal by continuously shifting the decision boundary. This observation motivates separating calibration from monitoring. In the proposed methodology, adaptation is restricted to a finite cold-start phase whose objective is to estimate an operational radius for each asset. Once convergence has been achieved, the shared representation remains fixed and subsequent departures become diagnostic evidence rather than input for further adaptation.

The present work is also related to invariant representation learning Bronstein et al. 2021 and anomaly detection under distribution shift **li2023invariant**, which seek representations that are insensitive to operationally irrelevant variability. Our contribution, however, lies at a different level of the condition-monitoring pipeline. Rather than proposing a new representation-learning architecture, we address the deployment problem itself: how a shared nominal representation can be deployed across heterogeneous assets through shared nominal coverage, independent operational-radius convergence, and local empirical calibration, without fault labels or per-asset retraining.

From a broader perspective, this deployment strategy follows the viewpoint that diagnosis should recover a canonical nominal representation from observations, while deployment should estimate only the local neighbourhood in which that representation remains valid for each individual asset. This interpretation is consistent with the inverse-problem and quotient-space formulations introduced in our previous work Di Santi 2026b; Di Santi 2026a, although the present paper focuses exclusively on the engineering problem of scalable fleet deployment.

The methodology is intentionally independent of both sensing modality and asset type. The present manuscript validates the approach on the CWRU bearing benchmark Smith and Randall 2015 because it provides controlled nominal observations, multiple fault modes, and several operating conditions. Bearings should therefore be interpreted as the first validation domain rather than as a limitation of the proposed deployment method.

Accordingly, this paper should be interpreted not as proposing another anomaly-detection architecture, but as introducing a deployment methodology in which a shared canonical nominal representation is combined with independent operational-radius convergence for each monitored asset.

# 3 Method

The objective of the proposed methodology is not to learn a detector for each asset independently, but to separate two distinct problems: learning a single shared canonical nominal representation for the fleet and estimating an asset-specific operational radius during deployment.

This section formalizes that deployment architecture.

## 3.1 Deployment architecture: shared representation, local radius

Let $x_{k,t} \in \mathcal{X}$ denote an observation window acquired from operational unit $k$ at time $t$. In the bearing validation presented in this paper, $x_{k,t}$ is instantiated as a fixed-length vibration window. A shared offline-trained representation model produces an anomaly score

$$r_{k,t} = r(x_{k,t}). \tag{1}$$

The representation model is common to the fleet. The decision boundary is not. Each asset maintains a local operational radius $\tau_k(t)$ and raises an alarm when

$$r_{k,t} > \tau_k(t). \tag{2}$$

The method therefore enforces three deployment rules:

1. the representation is shared;
2. the calibration is local;
3. the model is never retrained per asset.

**Shared offline nominal model**

$\downarrow$

| Asset A | Asset B | Asset C |
|---|---|---|
| $\tau_A(t) \to \tau_A^\star$ | $\tau_B(t) \to \tau_B^\star$ | $\tau_C(t) \to \tau_C^\star$ |

*Shared representation; independent operational-radius convergence.*

Figure 1: Fleet deployment architecture. One representation is distributed fleet-wide, while each physical asset converges independently to its own operational radius.

**Definition 1** (Operational radius)**.** *For a fixed nominal representation and an independently monitored operational unit $k$, the operational radius $\tau_k^\star$ is the converged, asset-specific acceptance boundary estimated from nominal deployment observations. It calibrates the anomaly decision locally without modifying or retraining the shared representation.*

## 3.2 Representation instantiation

The deployment method is independent of the representation backbone. In this paper, we instantiate it with a Conv1D autoencoder $A_\phi = D_\phi \circ E_\phi$ trained exclusively on nominal signals. The encoder maps a 1200-sample window through three Conv1D layers (32, 64, and 128 filters) to a 16-dimensional latent code with an $\ell_1$ activity penalty; the decoder mirrors this structure with Conv1DTranspose layers. The model has 847,601 parameters and is trained with Adam, mean-squared reconstruction loss, early stopping, and learning-rate decay.

The anomaly score is the reconstruction residual

$$r(x) = \|x - A_\phi(x)\|_2^2. \tag{3}$$

The autoencoder is merely one possible instantiation of the shared representation. The scientific contribution is the deployment methodology rather than the particular representation-learning architecture. Any nominal representation producing a stable scalar distance or anomaly score could replace Equation 3.

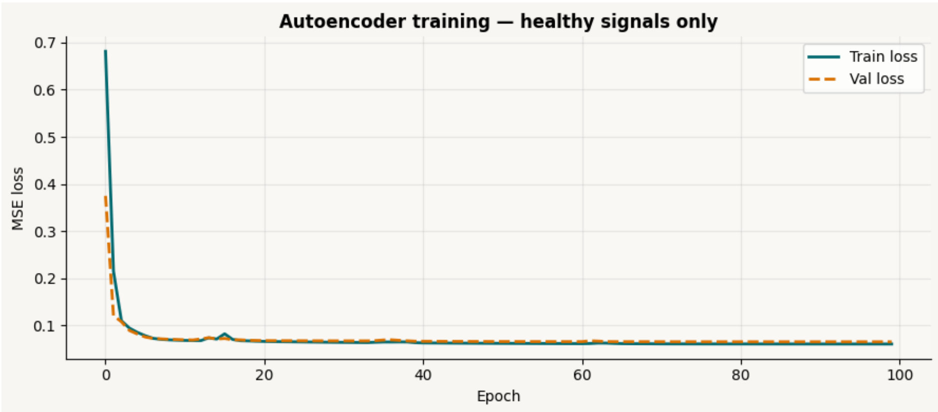


Figure 2: Training and validation loss of the healthy-only Conv1D autoencoder used as the shared nominal representation.

### 3.3 Shared nominal coverage and local empirical radius

The fleet-wide calibration parameter is a target nominal coverage level $1 - \alpha$, rather than an absolute anomaly-score threshold. In the present experiments, $1 - \alpha = 0.98$.

For operational asset $k$, let $\widehat{Q}_{k,t}(1 - \alpha)$ denote the empirical $(1 - \alpha)$-quantile of the local residuals observed during initialization up to time $t$. The local operational radius is

$$\tau_k(t) = \widehat{Q}_{k,t}(1 - \alpha). \tag{4}$$

The simulator threshold

$$\tau_{\mathrm{MC}} = \widehat{Q}_{\mathrm{MC}}(1 - \alpha) \tag{5}$$

is retained only for comparison. Its absolute numerical value is not used by the proposed field-deployment procedure. What is shared across the fleet is the target nominal coverage, while each asset identifies the residual value that realizes that coverage locally.

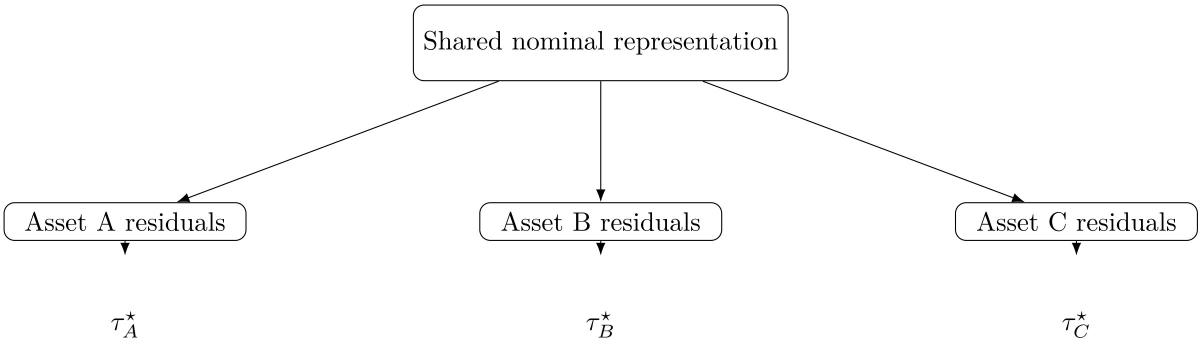


Figure 3: The same shared nominal representation produces reconstruction residuals for every deployed asset. Only the local empirical residual distribution changes across installations. Each asset independently estimates its own operational radius $\tau_k^\star$, while the representation remains unchanged.

### 3.4 Operational-radius convergence

During cold start, each asset collects a short sequence of initialization residuals and updates Equation 4. The asset is considered calibrated when the radius stabilizes.

For a look-back interval $L$, define the convergence statistic

$$C_k(t) = \frac{|\tau_k(t) - \tau_k(t-L)|}{\max\left(\tau_k(t-L), \epsilon\right)}, \tag{6}$$

where $\epsilon > 0$ prevents division by zero.

Calibration is declared when

$$C_k(t) < \varepsilon_C \tag{7}$$

for $m$ consecutive updates. The converged operational radius is then

$$\tau_k^\star = \tau_k(t_{\text{conv}}). \tag{8}$$

The key hypothesis evaluated in this paper is:

> *Under stable nominal operation, every asset converges rapidly to its own operational radius around a fixed shared representation.*

After convergence, $\tau_k^\star$ is frozen or allowed only a tightly bounded robust correction. This prevents a slowly increasing anomaly score from being absorbed as a new nominal state.

The convergence criterion therefore defines the transition from representation calibration to operational monitoring.

### 3.5 Guarded updates

For deployment beyond the controlled warm-up evaluated here, a guarded update rule can be applied to the local baseline. Let $\delta > 0$ define a guard band. A residual updates the local statistics only when

$$r_{k,t} < \tau_k(t) - \delta. \tag{9}$$

Samples inside the guard band are observed but do not update the baseline, and samples above $\tau_k(t)$ are flagged as anomalous. The three regions are therefore:

$$r_{k,t} < \tau_k(t) - \delta \quad \Rightarrow \quad \text{safe update}, \tag{10}$$

$$\tau_k(t) - \delta \le r_{k,t} \le \tau_k(t) \quad \Rightarrow \quad \text{observe, do not update}, \tag{11}$$

$$r_{k,t} > \tau_k(t) \quad \Rightarrow \quad \text{anomaly}. \tag{12}$$

This rule is deliberately lightweight: each asset stores only a residual buffer or running robust statistics, its current radius, and its convergence state.

### 3.6 Deployment states

Each asset transitions through three operational states:

1. **Cold start:** the target nominal coverage is converted into a local empirical operational radius from the incoming residual stream;
2. **Converged nominal:** $\tau_k^\star$ has stabilized and monitoring is active;
3. **Departure:** persistent scores exceed the converged radius.

A practical alarm may require $q$ exceedances within the last $p$ observations to avoid reacting to isolated impulses.

### 3.7 Scope of the Method

The deployment mechanism proposed in this paper is intentionally asset-agnostic. Throughout the paper, the index $k$ denotes an independently monitored operational unit. Depending on the application, $k$ may represent a rolling-element bearing, a traction motor, a gearbox, a pump, an actuator, a vehicle, or a spatially indexed infrastructure segment.

The methodology requires only: (i) a shared nominal representation, (ii) repeated observations from each operational unit, and (iii) local estimation of an operational radius.

The present manuscript validates the approach on the CWRU bearing benchmark because it provides controlled nominal observations, multiple fault modes, and several operating conditions. Bearings should therefore be interpreted as the first validation domain rather than as a limitation of the proposed deployment method.

## 4 Experimental Design

The experimental protocol is designed to validate the deployment hypothesis rather than the particular representation-learning architecture. Every stage, therefore measures some subset of the following quantities:

- false-alarm rate during cold start;
- number of accepted samples to operational-radius convergence;
- variability of $\tau_k^\star$ across assets and regimes;
- post-convergence detection rate;
- sensitivity to buffer size and convergence parameters;
- performance of absolute-threshold transfer versus shared-coverage calibration;
- cost of per-asset retraining avoided by the shared-model design.

**Stage 1 — Controlled multi-asset simulation.** Four bearing streams are generated with a physics-motivated nominal simulator using 1200-sample windows, a 30 Hz shaft frequency, and bounded variation in amplitude, phase, speed, offset, and noise. The shared model is trained on 4,000 simulated nominal windows, with 1,000 additional nominal windows reserved for validation. No simulated fault observations are used during representation learning. Faults are introduced only during evaluation. A separate synthetic test set contains 750 windows per regime. The simulator and all field experiments use fixed-length 1,200-sample windows. Each simulated asset begins in nominal operation and independently calibrates its radius. Asset A remains nominal; Assets B–D transition after convergence to outer-race, inner-race, and ball-fault regimes.

This stage measures: (i) samples to convergence; (ii) false alarms before and after convergence; (iii) the separation between $r_{k,t}$ and $\tau_k^\star$ after fault onset; and (iv) whether each nominal asset reaches a stable but asset-specific radius.

**Stage 2 — Direct simulation-to-field deployment on CWRU.** The nominal representation trained in Stage 1 exclusively on synthetic healthy signals is transferred unchanged to the Case Western Reserve University (CWRU) Drive-End data. No CWRU signal—healthy or faulty—is used to retrain, fine-tune, or otherwise modify the representation model. The purpose of this stage is to separate representation transfer from decision-boundary transfer.

Each CWRU operating condition is treated as an independent deployment environment. A short stream of unlabeled healthy observations is used only to estimate the empirical 98th-percentile residual and, consequently, the asset-specific operational radius $\tau_k(t)$. Fault observations are then evaluated using the unchanged simulator-trained model and the locally estimated decision boundary.

This stage tests two distinct hypotheses: (i) whether the nominal representation learned from simulation transfers directly to real field signals without retraining; and (ii) whether a shared target nominal coverage can be realized through a locally estimated empirical radius even though the absolute simulator threshold does not transfer.

The main outcome is not merely fault classification. It is whether each stream reaches a stable $\tau_k^\star$ during warm-up and then detects the subsequent departure without model retraining.

**Stage 3a — Ideal asset-specific reference.** To quantify the cost of scalability, we construct an intentionally favorable but operationally non-scalable reference condition. For each CWRU operating condition, three detectors are trained and calibrated independently using the complete available healthy dataset: (i) a Conv1D autoencoder; (ii) a one-class SVM operating on PCA-reduced standardized windows; and (iii) an Elliptic Envelope operating on PCA-reduced standardized windows.

Each detector uses the empirical 98th percentile of its own healthy anomaly-score distribution as the decision boundary. Fault observations are never used for model fitting or threshold selection. These experiments approximate the situation in which complete nominal data and asset-specific data-science support are available for every monitored unit.

**Stage 3b — Calibration-transfer ablation.** Using the unchanged simulator-trained autoencoder, we compare three deployment strategies:

1. **Direct threshold transfer:** apply the absolute synthetic p98 residual threshold unchanged to each CWRU condition;

2. **Field oracle:** estimate the p98 operational radius using the complete healthy field dataset of each condition, without retraining the shared representation; and

3. **Online local convergence:** estimate the p98 operational radius progressively from only the first 100 unlabeled healthy windows.

This ablation separates representation transfer from decision-boundary transfer. The representation is identical in all three conditions; only the information available for local calibration changes.

**Stage 4 — Multi-unit run-to-failure validation on NASA IMS.** A second nominal representation is constructed for the IMS bearing family using simulated healthy observations matched to the dataset sampling and operating characteristics. This model is trained once and then transferred unchanged to the independently monitored bearings within the IMS experiments. Each bearing estimates its own operational radius from an early-life unlabeled segment and subsequently enters post-convergence monitoring.

This stage evaluates whether the proposed deployment principle extends from heterogeneous CWRU operating conditions to physically distinct bearing units and naturally evolving degradation trajectories. We report: (i) samples to operational-radius convergence; (ii) variability of the converged radius across bearings; (iii) healthy-period false-alarm rate; (iv) first persistent departure from the frozen radius; and (v) alarm lead time relative to the documented end of the experiment.

**Stage 5 — Edge cost.** We report model size, parameter count, inference latency per 1200-sample window on CPU, local state memory, and the reduction in transmitted data when only scores and selected anomalous windows are sent rather than the continuous raw signal.

In the CWRU validation, the four rotational-speed conditions are treated as independent deployment environments rather than as four physically distinct fleet assets. They provide a controlled test of score-scale heterogeneity and local calibration, but do not reproduce the full manufacturing, installation, and aging variability of an operational fleet.

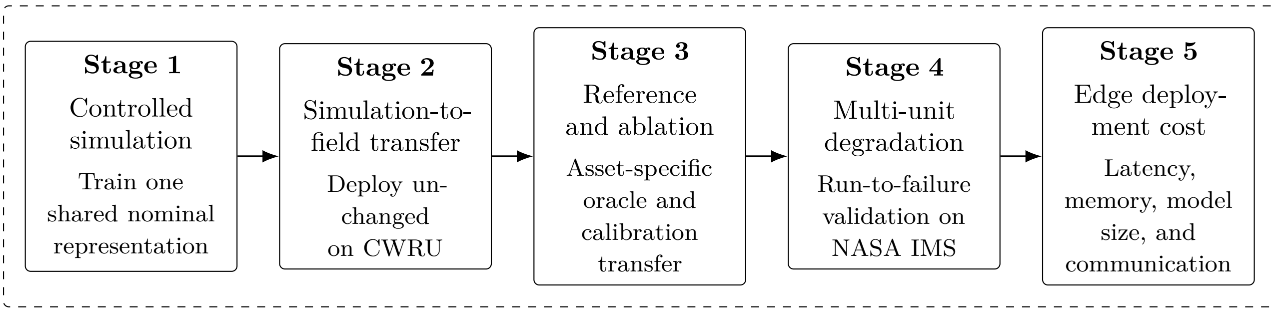


Figure 4: Deployment-oriented experimental protocol. The experiments progress from controlled representation learning to direct field transfer, ideal-reference comparison, multi-unit run-to-failure validation, and measurement of the edge-deployment footprint. Across deployment stages, the nominal representation remains fixed while each monitored unit estimates its own local operational radius.

# 5 Results

The experiments address two distinct questions. First, we establish the performance attainable under an ideal but operationally non-scalable condition in which a separate detector is trained and calibrated using the complete healthy dataset available for each CWRU operating condition. Second, we evaluate whether the proposed simulator-trained shared representation, combined only with a short online calibration stream, can approach that ideal performance without any field-model retraining.

The CWRU evaluation includes four operating conditions corresponding to 1730, 1750, 1772, and 1797 RPM. Across these conditions, the experiments contain 1,414 healthy windows, 1,621 ball-fault windows, 1,621 inner-race-fault windows, and 2,846 outer-race-fault windows. All windows contain 1,200 samples and are processed identically across detectors.

## 5.1 Ideal asset-specific reference

Table 1 reports the performance of three detectors trained separately for every CWRU operating condition using all available healthy observations from that condition. These experiments represent an oracle reference rather than a deployable fleet architecture: they assume that the complete nominal dataset has already been collected and that a data scientist can train, calibrate, and maintain one detector per operational condition.

For each detector, the decision boundary is the 98th percentile of its anomaly score on the corresponding healthy dataset. Consequently, the healthy false-alarm rate is approximately 2% by construction. The comparison therefore evaluates whether the detector score separates fault observations from the nominal observations, not whether its threshold can be calibrated.

Table 1: Ideal asset-specific detector performance on CWRU. Values are mean ± standard deviation across the four operating conditions. Each detector is trained and calibrated separately using the complete healthy dataset of the corresponding condition.

| Detector | Healthy FAR (%) | Ball DR (%) | Inner-race DR (%) | Outer-race DR (%) |
|---|---|---|---|---|
| Asset-specific AE | $2.29 \pm 0.12$ | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ |
| Asset-specific OC-SVM | $2.29 \pm 0.12$ | $3.09 \pm 6.17$ | $25.05 \pm 0.13$ | $2.71 \pm 3.15$ |
| Asset-specific Elliptic Envelope | $2.29 \pm 0.12$ | $0.62 \pm 1.23$ | $14.43 \pm 10.63$ | $1.76 \pm 3.15$ |

The asset-specific autoencoder separates every evaluated CWRU fault window from the calibrated nominal region, reaching 100% detection for all three fault families under all four operating conditions. In contrast, the classical one-class detectors satisfy the prescribed nominal false-alarm level but provide little diagnostic separation. The OC-SVM detects, on average, only 3.09% of ball faults, 25.05% of inner-race faults, and 2.71% of outer-race faults. The elliptical-envelope detector performs similarly, with average detection rates of 0.62%, 14.43%, and 1.76%, respectively.

These results show that nominal calibration alone is insufficient. All three methods reproduce the target healthy false-alarm level, but only the reconstruction-based representation produces an anomaly score that consistently separates the physical fault regimes. The asset-specific autoencoder is therefore used as the ideal performance reference for evaluating the cost of shared-model deployment.

## 5.2 Repeated-run stability of the asset-specific autoencoder

To evaluate the sensitivity of the asset-specific autoencoder to stochastic optimization, the complete training and evaluation procedure was repeated using five independent random initializations. The architecture, preprocessing, training protocol, datasets, and calibration procedure remained unchanged; only the random initialization and stochastic optimization trajectory differed between runs.

Table 2 summarizes the results. Across all operating conditions, no variation was observed in healthy false-alarm rate or fault detection performance. Every repeated run achieved identical detection rates for the three evaluated fault families, while maintaining the prescribed healthy false-alarm level.

The only quantity exhibiting noticeable variability was the absolute reconstruction-error threshold estimated from the healthy residual distribution. Depending on the operating condition, the threshold varied between repeated trainings despite producing identical detection performance.

Table 2: Repeated-run stability of the asset-specific autoencoder over five independent random initializations. Detection performance is unchanged across runs, whereas the absolute reconstruction threshold varies slightly.

| RPM | Healthy FAR (%) | Ball DR (%) | Inner DR (%) | Outer DR (%) | Threshold |
|---|---|---|---|---|---|
| 1730 | $2.23 \pm 0.00$ | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | $0.649 \pm 0.021$ |
| 1750 | $2.23 \pm 0.00$ | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | $0.759 \pm 0.227$ |
| 1772 | $2.23 \pm 0.00$ | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | $0.725 \pm 0.148$ |
| 1797 | $2.46 \pm 0.00$ | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | $0.459 \pm 0.026$ |

### 5.3 Direct simulation-to-field deployment

The shared autoencoder is trained once using simulated healthy observations and is transferred unchanged to all four CWRU conditions. No CWRU observation is used to retrain, fine-tune, or otherwise modify the model or its preprocessing transformation.

Three calibration conditions are evaluated:

1. **Direct threshold transfer**, in which the absolute p98 residual threshold estimated in simulation is applied unchanged in the field;
2. **Field oracle**, in which the p98 operational radius is computed using the complete healthy field dataset for each condition; and
3. **Proposed online convergence**, in which the local p98 operational radius is estimated progressively from only the first 100 unlabeled healthy windows.

Table 3 reports the complete results.

Table 3: Simulation-to-field deployment of the shared autoencoder on CWRU. FAR denotes the healthy false-alarm rate and DR the fault detection rate. The field-oracle radius uses every available healthy window; the online radius uses only the first 100 healthy windows. A dash indicates that convergence is not applicable because the threshold is computed directly rather than estimated sequentially.

| RPM | Calibration | $\tau$ | Healthy FAR (%) | Ball DR (%) | Inner DR (%) | Outer DR (%) | Healthy windows | Warm-up (windows) | Convergence (windows) |
|---|---|---|---|---|---|---|---|---|---|
| 1730 | Direct threshold transfer | 0.0979 | 100.00 | 100.00 | 100.00 | 100.00 | 404 | 0 | – |
| 1730 | Field oracle | 0.2005 | 2.23 | 100.00 | 100.00 | 100.00 | 404 | 404 | – |
| 1730 | Proposed online convergence | 0.1984 | 2.97 | 100.00 | 100.00 | 100.00 | 404 | 100 | 38 |
| 1750 | Direct threshold transfer | 0.0979 | 100.00 | 100.00 | 100.00 | 100.00 | 404 | 0 | – |
| 1750 | Field oracle | 0.1786 | 2.23 | 100.00 | 100.00 | 100.00 | 404 | 404 | – |
| 1750 | Proposed online convergence | 0.1759 | 2.97 | 100.00 | 100.00 | 100.00 | 404 | 100 | 58 |
| 1772 | Direct threshold transfer | 0.0979 | 100.00 | 100.00 | 100.00 | 100.00 | 403 | 0 | – |
| 1772 | Field oracle | 0.1997 | 2.23 | 100.00 | 100.00 | 100.00 | 403 | 403 | – |
| 1772 | Proposed online convergence | 0.2072 | 0.50 | 100.00 | 100.00 | 100.00 | 403 | 100 | 43 |
| 1797 | Direct threshold transfer | 0.0979 | 100.00 | 100.00 | 100.00 | 100.00 | 203 | 0 | – |
| 1797 | Field oracle | 0.2474 | 2.46 | 96.80 | 100.00 | 100.00 | 203 | 203 | – |
| 1797 | Proposed online convergence | 0.2364 | 5.91 | 98.77 | 100.00 | 100.00 | 203 | 100 | 65 |

Direct transfer of the absolute simulator threshold fails under every operating condition. Its 100% apparent fault-detection rate is meaningless because it simultaneously classifies 100% of the healthy field windows as anomalous. The shared representation therefore transfers, but its absolute reconstruction-error scale does not.

The field-oracle calibration restores the target false-alarm level, obtaining an average healthy FAR of $2.29 \pm 0.12\%$. With this complete knowledge of each field condition, the unchanged simulator-trained model obtains $99.20 \pm 1.60\%$ ball-fault detection and 100% detection of both inner- and outer-race faults.

The proposed online procedure reaches comparable performance using only the first 100 healthy windows. Across the four operating conditions, it achieves an average healthy FAR of $3.09 \pm 2.22\%$, ball-fault detection of $99.69 \pm 0.62\%$, and 100% detection of both inner- and outer-race faults. The online operational radius converges after $51.0 \pm 12.6$ windows, with observed convergence indices ranging from 38 to 65 windows.

## 5.4 Cost of scalable deployment

Table 4 compares the ideal asset-specific autoencoder with the proposed shared deployment architecture. The comparison is intentionally conservative in favor of the asset-specific reference. The oracle detector is trained and calibrated using the complete real healthy dataset for every operating condition. The proposed method receives no real field data for representation training and estimates its decision radius online from only 100 initial observations.

Table 4: Ideal but non-scalable asset-specific deployment versus the proposed fleet-scalable deployment. Values are mean ± standard deviation across the four CWRU operating conditions.

| Deployment strategy | Healthy FAR (%) | Ball DR (%) | Inner DR (%) | Outer DR (%) | Field model training | Healthy field data used |
|---|---|---|---|---|---|---|
| Asset-specific AE oracle | $2.29 \pm 0.12$ | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | One model per condition | Complete dataset |
| Shared AE with field-oracle radius | $2.29 \pm 0.12$ | $99.20 \pm 1.60$ | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | None | Complete dataset |
| Proposed shared AE with online radius | $3.09 \pm 2.22$ | $99.69 \pm 0.62$ | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | None | 100-window warm-up |

Replacing four separately trained real-data autoencoders with one simulator-trained shared model therefore produces no observed loss for inner- or outer-race detection and less than one percentage point of average difference for ball-fault detection. The principal cost appears in healthy false alarms, whose mean increases from 2.29% to 3.09% and whose variability is greater under the short online warm-up. This difference is concentrated primarily in the 1797 RPM condition, where the online FAR reaches 5.91%; the remaining conditions remain between 0.50% and 2.97%.

These results support the central deployment claim of this paper. Rather than demonstrating a new anomaly detector, they show that a single simulator-trained canonical representation can be deployed across heterogeneous assets with only a negligible reduction in detection performance, provided that each asset estimates its own local operational radius during a short unlabeled warm-up. This replaces asset-specific representation learning with lightweight local calibration, reducing deployment complexity while preserving near-native detection performance.

It is important to note that the comparison intentionally favors the asset-specific baselines. The baseline detectors are trained and calibrated using the complete healthy dataset available for each operating condition, whereas the proposed deployment estimates its operational radius from only the first 100 healthy windows (approximately ten seconds of operation) and never retrains the shared representation.

## 5.5 Operational-radius convergence

Figure 5 shows the sequential estimate of the local p98 operational radius during the 100-window warm-up. Each condition converges to a distinct radius despite using the same fixed representation. The final online estimates are close to their complete-data field-oracle counterparts: 0.1984 versus 0.2005 at 1730 RPM, 0.1759 versus 0.1786 at 1750 RPM, 0.2072 versus 0.1997 at 1772 RPM, and 0.2364 versus 0.2474 at 1797 RPM.

```
[PLACEHOLDER: operational-radius trajectories for 1730, 1750,
1772, and 1797 RPM, including the complete-data field-oracle
radius and detected convergence index.]
```

Figure 5: Online operational-radius convergence from the first 100 unlabeled healthy windows. The shared representation remains unchanged; only the local empirical p98 radius is estimated. Observed convergence occurs between 38 and 65 windows.

The convergence result is operationally relevant because the complete healthy distribution is not available at the start of a real deployment. The asset-specific and field-oracle references assume that all healthy data have already been collected. The proposed procedure instead constructs the local acceptance boundary incrementally and approaches the complete-data radius within a small fraction of the available nominal sequence.

The present experiment assumes that the initial warm-up observations are representative of nominal operation. Detecting an asset that is already degraded at installation, or determining that no stable nominal radius can be established, is outside the current scope. The convergence dynamics themselves may provide evidence of an invalid initialization state and are identified as a subject for future work.

### 5.6 Robustness to local initialization

A practical deployment question is whether the proposed local calibration depends critically on the particular nominal observations available during installation. In real deployments, the first healthy windows acquired from an asset are unlikely to be identical across installations, even when the underlying operating condition is the same.

To evaluate this sensitivity, the shared simulator-trained representation was kept completely fixed throughout the experiment. Neither the representation, its preprocessing, nor its parameters were modified. Only the local initialization data were varied.

For each CWRU operating condition, five independent warm-up sequences were generated by randomly sampling 100 healthy windows from the available field observations using different random seeds. Each warm-up sequence independently estimated the local empirical 98th-percentile operational radius, after which fault detection was evaluated using the unchanged shared representation.

Table 5: Robustness of online operational-radius estimation under different random warm-up samples. The shared simulator-trained representation remained fixed throughout the experiment. Only the 100 healthy initialization windows used for local calibration were randomly changed. Values are mean ± standard deviation over five random seeds.

| RPM | $\tau_{\text{online}}$ | Healthy FAR (%) | Ball DR (%) | Inner DR (%) | Outer DR (%) | Convergence (windows) |
|---|---|---|---|---|---|---|
| 1730 | $0.2905 \pm 0.0041$ | $4.08 \pm 2.04$ | $99.75 \pm 0.00$ | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | $45.6 \pm 13.2$ |
| 1750 | $0.3080 \pm 0.0063$ | $3.49 \pm 2.47$ | $99.95 \pm 0.11$ | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | $54.2 \pm 13.6$ |
| 1772 | $0.3097 \pm 0.0076$ | $2.18 \pm 1.66$ | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | $49.4 \pm 13.8$ |
| 1797 | $0.3415 \pm 0.0051$ | $3.49 \pm 2.44$ | $96.16 \pm 0.22$ | $100.00 \pm 0.00$ | $100.00 \pm 0.00$ | $56.2 \pm 15.0$ |

The estimated operational radii remained consistently close to the complete-data field-oracle values across all repetitions. Despite the variability introduced by different initialization samples, the online calibration systematically converged toward the same local operating region.

Detection performance was correspondingly stable. Ball-fault detection remained between approximately 96% and 100% across all evaluated seeds, while inner-race and outer-race faults continued to be detected with essentially unchanged performance. The principal variability appeared in the healthy false-alarm rate, which fluctuated depending on the particular warm-up sample used for estimating the empirical percentile.

This behaviour is expected from the statistical nature of the calibration procedure. Because the operational radius is estimated from only 100 initialization windows, the empirical 98th-percentile is determined by very few extreme observations. Small variations in those observations therefore produce modest variations in the estimated radius without materially affecting the diagnostic separation achieved by the shared representation.

These results indicate that the proposed deployment methodology is robust not only to simulation- to-field transfer but also to the specific nominal observations available during installation. The transferable object remains the shared canonical representation, whereas the local initialization procedure consistently identifies an appropriate operational radius from short, unlabeled field observations.

## 5.7 Edge deployment implications

The shared model contains 847,601 trainable parameters and is trained entirely offline. Deployment at the monitored asset requires only forward inference, a small residual buffer, estimation of an empirical quantile, and the current convergence state. No gradient computation, model optimization, fault labels, or asset-specific representation learning are required after installation.

On the evaluation platform, inference over a 1,200-sample window requires a median CPU latency of only 6.26 ms (95th percentile: 6.62 ms), substantially below the 100 ms acquisition interval corresponding to the evaluated windows. Consequently, the proposed deployment operates comfortably in real time while leaving ample computational margin for additional edge tasks such as communication, diagnostics, or signal preprocessing.

The local deployment state is also extremely compact. Besides the shared model itself, each asset stores only a buffer of the most recent 100 reconstruction residuals (400 bytes in float32) and a few scalar calibration variables, requiring less than 1 kB of asset-specific state. The only fleet-wide artifact is the shared nominal representation, which is trained once and distributed to every operational unit.

Table 6: Edge-deployment footprint of the proposed shared nominal representation.

| Metric | Value |
|---|---|
| Model parameters | 847,601 |
| Raw float32 weight size | 3.39 MB (3.23 MiB) |
| Serialized inference model (.keras) | 10.24 MB (9.76 MiB) |
| CPU inference latency (median) | 6.26 ms / 1,200-sample window |
| CPU inference latency (95th percentile) | 6.62 ms / window |
| Local residual-buffer length | 100 samples |
| Residual-buffer memory | 400 bytes |
| Total local calibration state | 428 bytes (< 1 kB) |
| Per-asset model training | None |
| Fault labels required | None |

These characteristics make the proposed methodology particularly suitable for edge deployment in large transportation fleets. Computational complexity remains essentially constant for each new asset because representation learning is centralized, whereas local deployment requires only inference and lightweight statistical estimation. Consequently, fleet growth increases the number of deployed instances but does not increase the number of models that must be engineered, validated, or maintained.

The measured latency corresponds to approximately 6% of the available processing budget for the evaluated acquisition rate (100 ms per window), providing roughly a $16\times$ real-time processing margin.

## 6 Discussion

The principal result of this work is not that a Conv1D autoencoder can separate CWRU fault classes. That capability has already been extensively demonstrated in the literature. The contribution lies instead in the deployment methodology: a single canonical nominal representation can be shared across heterogeneous assets, while each physical unit independently estimates its own operational radius without retraining the representation.

The repeated-run experiments provide an additional insight into this deployment strategy. Independent random initializations converge to slightly different numerical reconstruction-error scales, yet the resulting healthy false-alarm rate and fault detection performance remain essentially unchanged after empirical calibration. Different stochastic realizations therefore learn diagnostically equivalent nominal representations even though their absolute residual scales differ.

A complementary robustness experiment leads to the same conclusion from the deployment side. Keeping the representation fixed while varying only the nominal observations used during the 100-window warm-up produces only modest variations in the estimated operational radius and little change in post-calibration detection performance. The deployment mechanism is therefore robust at two independent levels: the learned representation itself and the local calibration procedure.

These observations reinforce the central claim of the paper. The transferable object is not an absolute reconstruction-error threshold but a shared canonical nominal representation. The role of local calibration is simply to identify the residual value that realizes the common target nominal coverage for each individual installation.

The calibration-transfer experiment further shows that the representation transfers successfully from simulation to field deployment, whereas the absolute reconstruction-error scale does not. A fixed simulator threshold therefore cannot be deployed fleet-wide. What transfers is the representation together with the target nominal coverage, while each asset estimates its own local operational radius from its residual distribution.

This deployment methodology also fits naturally within the broader theoretical framework developed in our previous work. In our earlier technical report Di Santi 2026b, industrial diagnosis was interpreted as recovering a canonical nominal representation from observations rather than directly classifying raw signals. That interpretation was subsequently formalized as an inverse problem over quotient spaces Di Santi 2026a, where the canonical representation constitutes the mathematical object to be recovered while nuisance variability is factored out.

The present work addresses the deployment stage of that formulation. Once the shared canonical representation has been learned offline, each deployed asset no longer needs to solve the global representation-learning problem. Instead, deployment consists only of solving a lightweight inverse calibration problem: given the reconstruction residuals generated by the fixed shared representation, each asset estimates the operational radius that best characterizes its own nominal operating conditions.

This viewpoint is also consistent with the blueprint formulation introduced in Di Santi 2025, where diagnosis was interpreted as recovering a compact nominal structure from observations. Here, that recovered canonical representation is assumed to be available, and only its local realization must be identified for each deployed asset. Consequently, representation learning solves the global inverse problem only once, whereas deployment repeatedly solves a lightweight local inverse calibration problem for every physical asset.

Operational-radius convergence also clarifies the role of adaptation. The objective of adaptation is to identify the asset during cold start, not to follow its aging indefinitely. Once $\tau_k^\star$ has been established, continued movement of the anomaly-score distribution constitutes diagnostic evidence. Allowing unrestricted adaptation after convergence would instead risk absorbing progressive degradation into the nominal baseline.

The observed variability in healthy false-alarm rate should be interpreted in the context of the deliberately short initialization interval. Using 1,200-sample non-overlapping windows acquired at 12 kHz, the 100-window warm-up corresponds to only approximately ten seconds of operation. Estimating an empirical 98th percentile from such a short sequence inevitably depends on a very small number of extreme observations.

A practical deployment would not necessarily terminate initialization after a fixed number of windows. Instead, the operational radius could continue to evolve automatically until the convergence criterion is satisfied, or until a minimum amount of nominal operating time has been observed. Once convergence is achieved, the radius can be frozen or tightly constrained so that subsequent degradation cannot redefine itself as nominal.

From an industrial perspective, the proposed architecture offers several practical advantages. It enables day-one deployment without fault labels, removes per-asset representation learning, isolates calibration between physical assets, requires only lightweight edge computation, and reduces continuous transmission of high-frequency sensor data.

An additional consequence is that computational scalability becomes decoupled from fleet size. Fleet growth increases the number of monitored assets, but not the number of engineered models. Adding a new operational unit requires only distributing the shared representation and estimating its local operational radius. Representation engineering, validation, and maintenance remain constant regardless of the number of deployed assets.

The computational requirements further support this deployment model. On the evaluation platform, inference requires only 6.26 ms for a 1,200-sample window, corresponding to approximately 6% of the available processing budget for the evaluated acquisition rate (100 ms per window). The system therefore provides roughly a sixteen-fold real-time processing margin while requiring less than 1 kB of asset-specific deployment state beyond the shared representation.

An additional practical advantage is that calibration is performed around a previously learned canonical nominal representation rather than around the deployment data themselves. Asset-specific retraining implicitly assumes that the calibration dataset is entirely healthy. If the monitored asset already exhibits incipient degradation, that degradation may become part of the learned nominal model. In contrast, the proposed methodology preserves the shared canonical representation and estimates only a local operational radius. This separation opens the possibility of identifying assets whose initial behavior is already inconsistent with the canonical nominal manifold before they are accepted as locally healthy.

## 6.1 Future Work

The present work intentionally focuses on day-one deployment under the assumption that the monitored asset begins operation in a nominal condition. A natural extension is to relax this assumption.

Because the methodology separates the shared canonical representation from the asset-specific operational radius, future work will investigate whether the initialization phase itself can be validated against the canonical nominal manifold. Such a consistency test could distinguish genuinely healthy initialization from assets already exhibiting incipient degradation, thereby preventing abnormal behavior from becoming part of the local baseline.

A second research direction concerns progressive physical degradation. Rather than detecting only departures beyond a converged operational radius, future work will investigate whether the joint evolution of reconstruction residuals and operational-radius dynamics contains predictive information about the degradation process. This would naturally connect the proposed deployment methodology with physics-informed health indicators and remaining useful life estimation.

Finally, the mathematical interpretation introduced in our previous work Di Santi 2026a opens several theoretical questions. If the shared canonical nominal representation is indeed the transferable mathematical object, then the operational radius may be interpreted as defining an asset-specific neighborhood around that object. Formalizing this relationship, characterizing its invariance properties, and establishing conditions under which local calibration preserves the underlying quotient-space structure constitute promising directions for future research.

# 7 Conclusion

We presented a day-one deployment methodology for nominal condition monitoring across heterogeneous asset fleets, validated on public rolling-element bearing datasets. A single nominal representation is trained once on simulated healthy observations and transferred unchanged to field deployment. Each operational unit then performs only a short unlabeled warm-up to estimate its own operational radius, without modifying or retraining the shared representation.

The experimental results show that a simulator-trained nominal representation transfers successfully to field deployment, whereas an absolute decision threshold does not. A shared target nominal coverage, realized locally through the empirical residual quantile estimated during initialization, provides an effective bridge between the shared representation and field deployment. This deployment strategy preserves detection performance close to that of asset-specific models trained under ideal asset-specific calibration conditions while eliminating per-asset representation learning.

The transferable object is therefore not an absolute anomaly threshold, nor an asset-specific representation, but a shared canonical nominal representation together with a lightweight local calibration mechanism. The central outcome of this work is consequently not a new anomaly detector, but a scalable deployment principle: learn the canonical nominal representation once, distribute it across the fleet, and let each physical asset identify only its own operational radius during installation.

This separation between representation learning and deployment calibration provides a practical foundation for large-scale edge condition monitoring in future transportation systems.

Such a representation could naturally be produced and maintained by the original equipment manufacturer (OEM), while fleet operators perform only lightweight local calibration during deployment, thereby separating representation engineering from operational deployment.

# 8 Compact Mathematical Perspective

The deployment methodology presented in this paper is intentionally engineering-oriented. Its mathematical foundations are developed in our accompanying technical reports on inverse recovery and quotient-space formulations Di Santi 2026b; Di Santi 2026a. The purpose of this section is not to reproduce those developments, but to summarize the key mathematical ideas that justify why a shared canonical nominal representation and independently estimated operational radii are compatible.

## 8.1 Observation model

Let $\mathcal{X} \subset \mathbb{R}^n$ denote the space of fixed-length observation windows, $\theta \in \Theta$ the latent physical state, and $\eta \in \mathcal{E}$ the nuisance variables. The observation process is

$$x = F(\theta, \eta). \tag{13}$$

Let $G_{\text{nuis}}$ denote a physically motivated family of nuisance transformations that preserve the underlying diagnostic state. Two observations are therefore considered equivalent whenever they differ only through such nuisance transformations.

### 8.2 Existence of a compact nominal representation

**Proposition 1** (Relative compactness of the nominal orbit). *If $\mathcal{E}$ is compact and $\eta \mapsto F(\theta_{\mathrm{N}}, \eta)$ is uniformly bounded and equicontinuous in the sup-norm topology, then the nominal orbit*

$$\mathcal{O}_{\mathrm{N}} = \{F(\theta_{\mathrm{N}}, \eta) : \eta \in \mathcal{E}\}$$

*is relatively compact.*

*Proof.* The result follows directly from the Arzelà–Ascoli theorem. Consequently, its closure $\overline{\mathcal{O}_{\mathrm{N}}}$ is compact. □

A representation that collapses nuisance-equivalent observations may therefore map nominal operation into a compact canonical structure. The shared representation learned offline approximates this structure, whereas the operational radius accounts only for the asset-specific numerical scale and dispersion of the anomaly score. Estimating $\tau_k^\star$ therefore does not modify the representation itself; it identifies the local acceptance neighborhood around the same canonical nominal object.

**Proposition 2** (Factorization of invariant diagnostic maps). *If a diagnostic map is invariant under the action of $G_{\mathrm{nuis}}$, then it is constant on nuisance-equivalence classes and factors through the quotient projection.*

This proposition provides the geometric justification for treating diagnosis as membership testing around a canonical nominal representation rather than directly on raw observations. Within this interpretation, representation learning solves the global inverse problem only once, whereas deployment solves a lightweight local inverse calibration problem by estimating the operational radius associated with each physical asset.

The present paper validates experimentally this deployment principle. A complete mathematical treatment of the inverse-problem formulation, quotient-space construction, and their relation to canonical nominal representations is provided in Di Santi 2026b; Di Santi 2026a.

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